

JARO 2025-2026

Assignment-1 Linear Regression

Introduction

In this assignment, you will implement a Linear Regression model from scratch using only NumPy and Pandas. You will explore the impact of learning rates, analyze model generalization, and implement regularization techniques ($L1$ and $L2$) to understand their effect on model weights.

1. Data Acquisition

Use the following Python snippet to download and load the dataset:

```
import kagglehub
import pandas as pd

# Download latest version
path = kagglehub.dataset_download("yasserh/housing-prices-dataset")
dataset = pd.read_csv(f'{path}/Housing.csv')
```

2. Demo Notebook

You can use the following demo notebook as a starting point for your implementation:

<https://drive.google.com/file/d/1ztAN90MNd06In44eFLRBItVPG124aEoE/view?usp=sharing>

3. Assignment Questions

1. **Implementation:** Implement a Linear Regression model from scratch. Use Mean Squared Error (MSE) as your loss function:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - (\mathbf{w} \cdot \mathbf{x}_i + b))^2$$

2. **Gradient Derivation:** Compute the gradients for both the weights (w) and the bias (b) mathematically and implement them in your code.
3. **Learning Rate Analysis:** Train your model using the following learning rates: $\{10^{-4}, 10^{-2}, 0.1, 1, 5\}$.
 - Plot and observe the loss curves for each.
 - Which learning rate performed best? Provide a justification based on your observations.
4. **Evaluation Metrics:** Which metric(s) should be used to analyze the model performance? Explain the rationale behind your choice.

5. **Generalization:** Evaluate the model on both training and validation sets. Compare the metrics and discuss whether the model is generalizing well or if it is overfitting/underfitting.
6. **L1 Regularization (Lasso):** Implement $L1$ regularization. Train the model with varying weight decay (λ) values: $\{10^{-9}, 10^{-6}, 10^{-3}, 1, 5\}$.
7. **L2 Regularization (Ridge):** Implement $L2$ regularization. Train the model with the same weight decay values: $\{10^{-9}, 10^{-6}, 10^{-3}, 1, 5\}$.
8. **Weight Analysis:** Examine the weights of the model across different iterations for both $L1$ and $L2$.
 - What can you infer about the magnitude of the weights?
 - What specific effect do $L1$ and $L2$ have on the weight vector?
9. **Comparison:** Compare the results of $L1$ and $L2$ regularization. In the context of this housing dataset, which one yielded a better-performing model?
10. **Final Insights:** Is the model generalizing better after applying regularization? Provide your insights supported by evidence from your training/validation metrics.